

JEE-Main-07-04-2025 (Memory Based)**[EVENING SHIFT]****Physics**

Question: Given below are two statements. One is labelled as Assertion (A) and the other is labelled as reason (R).

Assertion (A): Refractive index of glass is more than air.

Reason (R): Optical density of a medium is directly related to its mass density.

In the light of the above statements, choose the correct answer from the options given below

Options:

- (a) (A) is false but (R) is true
- (b) (A) is true but (R) is false
- (c) (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (d) (A) and (R) are true and (R) is the correct explanation of (A)

Answer: (b)

Question: The figure shows a circular portion of radius $R/2$ removed from a disc of mass m and radius R . The moment of inertia about an axis passing through the centre of mass of the disc and perpendicular to the plane is



Options:

- (a) $\frac{13}{32}mR^2$
- (b) $\frac{mR^2}{2}$
- (c) $\frac{mR^2}{4}$
- (d) $\frac{13}{64}mR^2$

Answer: (a)

Question: Give below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): A magnetic monopole does not exist.

Reason (R): Magnetic lines are continuous and form closed loops.

In the light of the above statements, choose the correct answer from the options given below:

Options:

- (a) (A) is false but (R) is true
- (b) (A) is true but (R) is false
- (c) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- (d) Both (A) and (R) are true and (R) is the correct explanation of (A)

Answer: (d)

Question: Potential energy is not defined for which of the force

Options:

- (a) Gravitational force
- (b) Restoring force
- (c) Friction
- (d) Electrostatic force

Answer: (c)

Question: Which of the following quantity has same dimensions as $\sqrt{\frac{\mu_0}{\epsilon_0}}$?

Options:

- (a) Voltage
- (b) Resistance
- (c) Inductance
- (d) Capacitance

Answer: (b)

Question: The equation of a wave travelling on a string is $y = \sin 20\pi t + 10\pi t$ where x and t are distances and time in SI units, the minimum distance two points having same oscillating speed is

Options:

- (a) 10 cm
- (b) 20 cm
- (c) 5.0 cm
- (d) 2.5 cm

Answer: (c)

Question: An object with mass 500 g moves along x-axis with speed $V = 4\sqrt{x} \text{ m/s}$ then force acting on the object is

Options:

- (a) 8 N
- (b) 6 N
- (c) 5 N
- (d) 4 N

Answer: (d)

Question: Find ratio of average kinetic energy of equal mass of He and Ar gas at 300 K.

Options:

- (a) 10 : 1

- (b) $1 : \sqrt{10}$
- (c) $\sqrt{10} : 1$
- (d) $2 : \sqrt{5}$

Answer: (a)

Question: The unit of $\sqrt{\frac{2I}{\epsilon_0 c}}$ is _____ where I = intensity of an electro magnetic wave, c = speed of light

Options:

- (a) NC
- (b) NC^{-1}
- (c) Nm
- (d) Vm

Answer: (b)

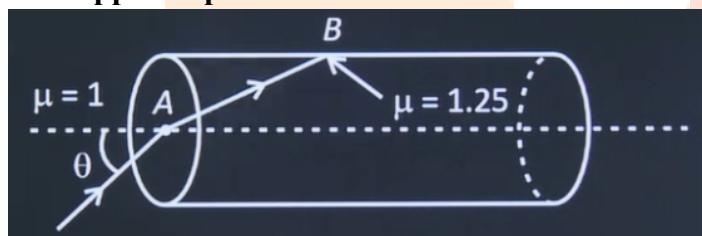
Question: An object is released from a plane moving horizontally with a speed 100 m/s at a height 2 km above ground. The horizontal distance travelled (in km) by the object is (take $g = 10 \text{ m/s}^2$)

Options:

- (a) 0.5
- (b) 2
- (c) 1
- (d) 2.5

Answer: (b)

Question: The maximum value of θ (shown in figure) for which total internal reflection can happen at point B is



Options:

- (a) $\tan^{-1}\left(\frac{4}{3}\right)$
- (b) $\sin^{-1}\left(\frac{3}{4}\right)$
- (c) $\cot^{-1}\left(\frac{3}{4}\right)$
- (d) $\cos^{-1}\left(\frac{3}{4}\right)$

Answer: (b)

Question: Distance between object and image for a convex lens is 40 cm and

magnification is $-\frac{1}{4}$. Find focal length of the lens.

Options:

- (a) 14.5 cm
- (b) 15 cm
- (c) 12.5 cm
- (d) 6.4 cm

Answer: (d)

Question: Flux through a plane parallel to x-z plane is 6 SI units. Find area of plane if

electric field in the region is $\vec{E} = (\hat{i} + 4\hat{j} + \hat{k})10^3 \text{ N/C}$.

Options:

- (a) $2 \times 10^{-3} \text{ m}^2$
- (b) $2.5 \times 10^{-2} \text{ m}^2$
- (c) $1.5 \times 10^{-3} \text{ m}^2$
- (d) $2.5 \times 10^{-3} \text{ m}^2$

Answer: (c)

Question: Match the List-I with the List-II.

	List-I		List-II
i	Isothermal	a	$\Delta W = 0$
ii	Adiabatic	b	$\Delta Q = 0$
iii	Isobaric	c	$\Delta U \neq 0$
iv	Isochoric	d	$\Delta U = 0$

Options:

- (a) (i)-(a), (ii)-(b), (iii)-(d), (iv)-(c)
- (b) (i)-(d), (ii)-(b), (iii)-(c), (iv)-(a)
- (c) (i)-(d), (ii)-(a), (iii)-(c), (iv)-(b)
- (d) (i)-(c), (ii)-(b), (iii)-(d), (iv)-(a)

Answer: (b)

Question: Match the List-I with the List-II.

	List-I		List-II
i	Mass density	a	$[\text{ML}^2\text{T}^{-3}]$
ii	Impulse	b	$[\text{MLT}^{-1}]$

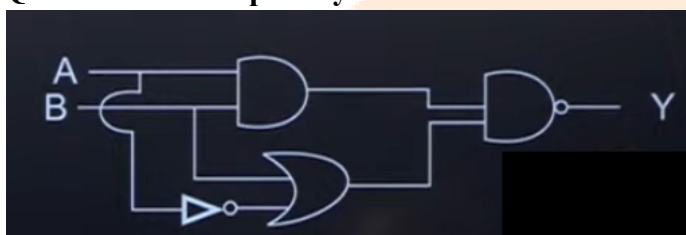
ii i	Power	c	$[ML^2T^0]$
iv	Moment of inertia	d	$[ML^{-3}T^0]$

Options:

- (a) (i)-(a), (ii)-(c), (iii)-(d), (iv)-(b)
- (b) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)
- (c) (i)-(d), (ii)-(a), (iii)-(c), (iv)-(b)
- (d) (i)-(d), (ii)-(b), (iii)-(a), (iv)-(c)

Answer: (d)

Question: The output is $y = 0$ when



Options:

- (a) $A = 0, B = 1$
- (b) $A = 1, B = 0$
- (c) $A = 1, B = 1$
- (d) $A = 0, B = 0$

Answer: (c)

Question: A photon of wavelength λ_1 is incident on a photo-sensitive surface of threshold wavelength λ_0 . The de-Broglie wavelength of fastest photoelectron is

Options:

- (a) $\sqrt{\frac{h}{2mc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_0} \right)}}$
- (b) $\sqrt{\frac{h}{2mc} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_0} \right)}$
- (c) $\lambda_1 - \lambda_0$
- (d) $\sqrt{\lambda_1^2 - \lambda_0^2}$

Answer: (a)

Question: Assertion: Airplane is made of metal to prevent from lightning strike.

Reason: Electric field in cavity inside conductor at equilibrium remains zero.

Options:

- (a) Both Assertion and Reason are correct
- (b) Assertion is correct but Reason is incorrect
- (c) Assertion is incorrect and Reason is correct

(d) Both are incorrect

Answer: (a)

Question: Charge on capacitor plate is $5 \times 10^{-6} \text{C}$ and induced charge on dielectric slab is $4 \times 10^{-6} \text{C}$. Find dielectric constant of the slab.



Options:

(a) 2

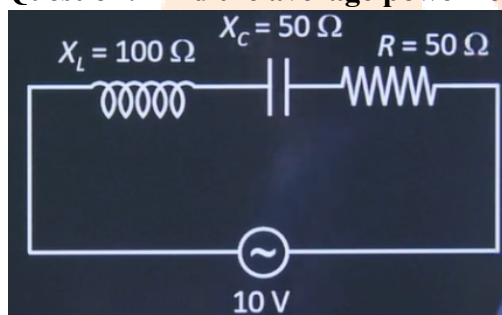
(b) 3

(c) 4

(d) 5

Answer: (d)

Question: Find the average power for the given AC circuit



Options:

(a) 1 W

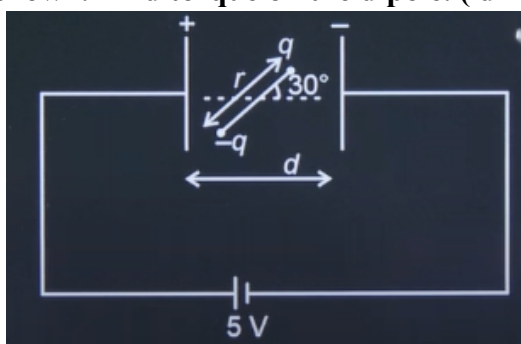
(b) 2 W

(c) 0.5 W

(d) 4 W

Answer: (a)

Question: An electric field dipole is kept between plates of a parallel plate capacitor as shown. Find torque on the dipole. ($d = 1 \text{mm}$, $q = 2 \mu\text{C}$, $r = 0.5 \mu\text{m}$)



Options:

- (a) $5 \times 10^{-9} \text{ Nm}$
- (b) $2.5 \times 10^{-9} \text{ Nm}$
- (c) $5 \times 10^{-12} \text{ Nm}$
- (d) $2 \times 10^{-8} \text{ Nm}$

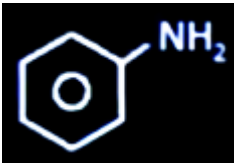
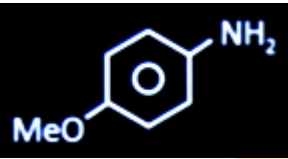
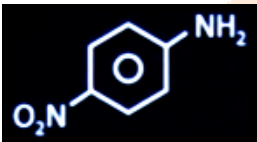
Answer: (b)



JEE-Main-07-04-2025 (Memory Based)
[EVENING SHIFT]

Chemistry

Question: The correct order to basic strength of the following molecules is

- (A) 
- (B) 
- (C) 

(D) CH_3NH_2

Options:

- (a) $A > B > C > D$
(b) $B > C > D > A$
(c) $D > B > A > C$
(d) $B > A > C > D$

Answer: (c)

Question: Which of the following is the correct Hybridisation of XeF_4 ?

Options:

- (a) sp^3d
(b) sp^3
(c) sp^3d^2
(d) sp^3d^3

Answer: (c)

Question: Which of the following is correct order of acidic character of oxides of vanadium?

Options:

- (a) $\text{V}_2\text{O}_5 > \text{VO}_2 > \text{V}_2\text{O}_3$
(b) $\text{V}_2\text{O}_3 > \text{VO}_2 > \text{V}_2\text{O}_5$
(c) $\text{V}_2\text{O}_5 > \text{V}_2\text{O}_3 > \text{VO}_3$
(d) $\text{VO}_2 > \text{V}_2\text{O}_3 > \text{V}_2\text{O}_5$

Answer: (a)

Question: If 174 g of Butane reacts with 320 g of O_2 . Find the volume of H_2O formed in ml.

(given density of H_2O is 1 g/ml)

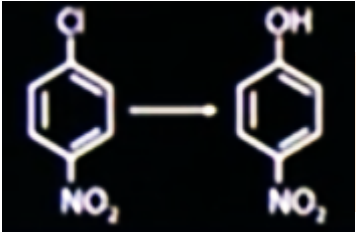
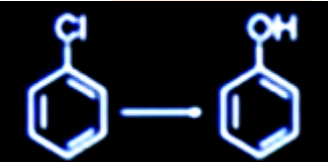
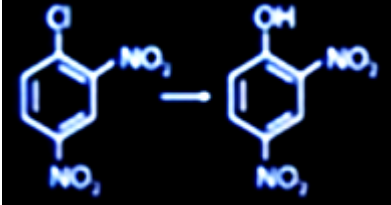
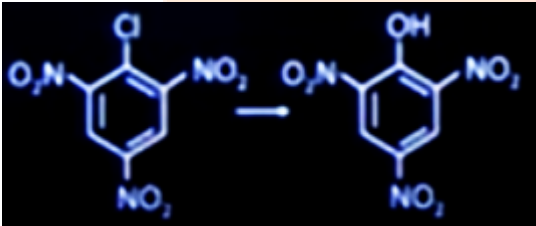
Options:

- (a) (i) sp^2 (ii) sp^2 (iii) sp^2

- (b) (i) sp^2 (ii) sp (iii) sp^2
(c) (i) sp^3 (ii) sp (iii) sp^2
(d) (i) sp (ii) sp^2 (iii) sp^3

Answer: (b)

Question: Match the following List-I with List-II

List-I (Reactions)	List-II (Reaction Temperature)
(A) 	Slight Warming
(B) 	368 K
(C) 	443 K
(D) 	623 K

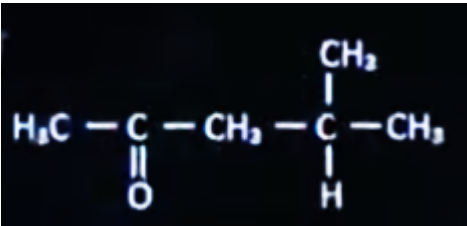
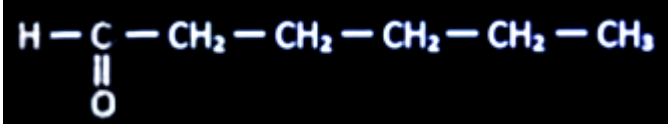
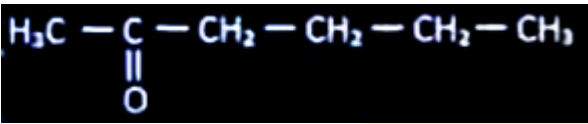
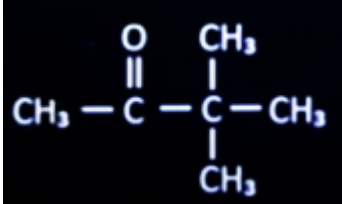
Options:

- (a) A-II, B-III, C-I, D-IV
(b) A-IV, B-III, C-II, D-I
(c) A-I, B-II, C-III, D-IV
(d) A-II, B-IV, C-III, D-I

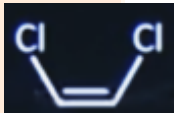
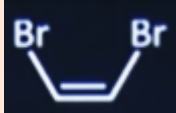
Answer: (b)

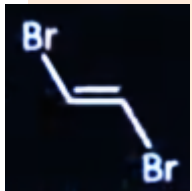
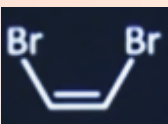
Question: Which of the following compound molecular formula $C_6H_{12}O$. Positively to 2, 4-DNP test and Tollen's reagent test

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (b)

Question: Assertion:  has more dipole moment than 

Reason:  has more boiling point than 

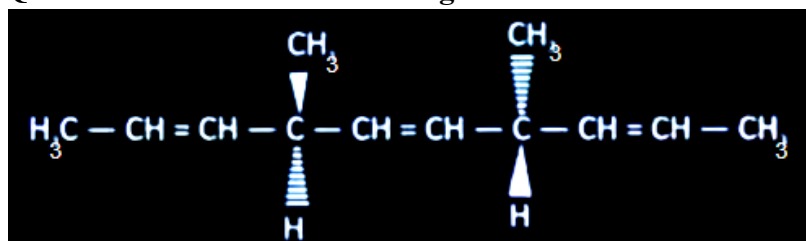
Choose the correct option.

Options:

- (a) Both Assertion & Reason are correct, Reason is correct explanation of Assertion
 (b) Both Assertion & Reason are correct, Reason is not correct explanation of Assertion
 (c) Assertion is correct, Reason is incorrect
 (d) Assertion is incorrect, Reason is correct

Answer: (c)

Question: Consider the following molecule



Number of optically active molecule(s) formed after complete reductive ozonolysis of above compound is

Options:

- (a) 2
(b) 1
(c) 0
(d) 3

Answer: (c)

Question: If $P_A^\circ = 350$ torr and $P_B^\circ = 750$ torr and the two volatile liquids (A) and (B) form an ideal solution. X_A and X_B are the respective mole fraction of (A) and (B) in solution and Y_A and Y_B are the respective mole fractions of (A) and (B) in the vapour phase. What one of the following relation is correct?

Options:

- (a) $\frac{Y_A}{Y_B} = \frac{X_A}{X_B}$
(b) $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$
(c) $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$
(d) $\frac{Y_A}{Y_B} = \frac{X_B}{X_A}$

Answer: (b)

Question: Among the given order, the incorrect order of atomic radii is

Options:

- (a) $r_{Rb} < r_{Cs}$
(b) $r_{Mg} < r_{Al}$
(c) $r_{Cl} < r_{Br}$
(d) $r_K < r_{Rb}$

Answer: (b)

Question: Match List-I with List-II and select the correct option

List-I (Solution)	List-II (Properties)
(A) Benzene + Toluene	(P) Show +ve deviation
(B) Aniline + CH_3COOH	(Q) $\Delta V_{\text{mix}} = 0$
(C) Water + ethanol	(R) $\Delta H_{\text{mix}} = -ve$
(D) Acetone + CHCl_3	(S) form minimum boiling azeotrope

Options:

- (a) A-Q, B-R, C-P, S D-R
(b) A-S, B-Q, C-P, S D-R
(c) A-Q, B-P, S, C-R, D-P
(d) A-P, S, B-S, C-P, D-R

Answer: (a)

Question: List-I mentions thermodynamic process of list-II mention property

List-I	List-II
(A) Isothermal	I. $Q = 0$
(B) Adiabatic	II. $\Delta T = 0$
(C) Isobaric	III. $\Delta T = 0$
(D) Isobaric	IV. $\Delta P = 0$

Options:

- (a) A-II, B-I, C-IV, D-III
- (b) A-I, B-II, C-III D-IV
- (c) A-I, B-IV, C-III, D-II
- (d) A-II, B-I, C-III, D-IV

Answer: (a)

Question: In electrolysis of aqueous solution of AgNO_3 , $\text{Cu(NO}_3)_2$, $\text{Hg(NO}_3)_2$ and $\text{Au(NO}_3)_3$ is carried out, then correct order of deposition of metal cathode give

Metal ion	SRP(V)
Ag^+/Ag	0.79
Cu^{2+}/Cu	0.34
Hg^{2+}/Hg	0.85
Au^{2+}/Au	1.4

Options:

- (a) $\text{Au} > \text{Hg} > \text{Ag} > \text{Cu}$
- (b) $\text{Au} > \text{Hg} > \text{Cu} > \text{Ag}$
- (c) $\text{Au} > \text{Ag} > \text{Hg} > \text{Cu}$
- (d) $\text{Cu} > \text{Ag} > \text{Hg} > \text{Au}$

Answer: (a)

Question: Hydration energy of K^+ is $-x$ kJ/mol and of Cl^- is $-y$ kJ/mol and lattice energy of KCl is $-Z$ kJ/mol then what is the heat of dissolution of KCl?

Options:

- (a) $Z - (x + y)$
- (b) $-Z - (x + y)$
- (c) $Z + x + y$
- (d) $-Z + (x + y)$

Answer: (a)

Question: How many of the following coordination compounds having same coordination number and paramagnetic in nature?

$\text{K}_3[\text{Fe(CN)}_6]$, $\text{K}_4[\text{Fe(CN)}_6]$, $[\text{Cu(NH}_3)_4]\text{SO}_4$,
 K_2HgI_4 , $[\text{Ni(H}_2\text{O)}_6]^{2+}$

Options:

- (a) 2
- (b) 3
- (c) 1

(d) 4

Answer: (a)

Question: Choose correct option for Reducing Nature of Stability of oxidation in following.

Options:

(a) Stability $Ti^{+1} < Ti^{+3}$

(b) Reducing nature $Ti^{+1} > Ti^{+3}$

(c) Stability $Al^{+} > Al^{3+}$

(d) Reducing nature $Al^{+1} > Al^{+3}$

Answer: (d)

Question: Choose correct option for Reducing Nature of Stability of oxidation in following.

Options:

(a) Stability $Ti^{+1} < Ti^{+3}$

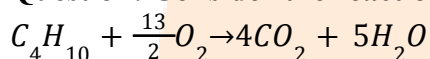
(b) Reducing nature $Ti^{+1} > Ti^{+3}$

(c) Stability $Al^{+} > Al^{3+}$

(d) Reducing nature $Al^{+1} > Al^{+3}$

Answer: (d)

Question: Consider the reaction given below:



If 174 g of Butane reacts with 320 g of O_2 . Find the volume of H_2O formed in ml.
(given density of H_2O is 1 g/ml)

Answer: (138.46)

Question: Consider the following oxides

V_2O_5 , Cr_2O_3 , Mn_2O_7 , V_2O_3 , VO_2

Number of oxides which are acidic is x.

Consider the following complex compound $[Co(NH_2CH_2CH_2NH_2)_3]_2(SO_4)_3$ the primary valency of complex is y

The value of x + y is

Answer: (4)

JEE-Main-07-04-2025 (Memory Based)
[EVENING SHIFT]
Maths

Question: If $x|x - 3| + 3|x - 2| + 1 = 0$, then the number of real solution is

Options:

- (a) 2
- (b) 4
- (c) 1
- (d) 6

Answer: (1)

Question: $\operatorname{Re} \left(\frac{2z+i}{z+i} \right) + \operatorname{Re} \left(\frac{2\bar{z}-i}{\bar{z}-i} \right) = 2$ is a circle of radius r and centre (a, b) ,
 then $\frac{15ab}{r^2}$ is equal to

Answer: (0)

Question: If two vectors $\vec{a}$ and $\vec{b}$ satisfies $\frac{|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}|} = \sqrt{2} + 1$, then the value of $\frac{|\vec{a} + \vec{b}|^2}{|\vec{a} - \vec{b}|^2}$ is equal to

Options:

- (a) $1 + \sqrt{2}$
- (b) $2 + 4\sqrt{2}$
- (c) $1 + 2\sqrt{2}$
- (d) $3 + 2\sqrt{2}$

Answer: (d)

Question: Let $f(x) = \frac{x-5}{x^2-3x+2}$ is range of $f(x)$ is $(-\infty, \alpha) \cup (\beta, \infty)$. Then $\alpha^2 + \beta^2$ equals to

Answer: (194)

Question: If a box contains 19 unbiased coins and 1 biased coin with both faces head. If a coin is randomly chosen out of this box and tossed. If the head appears, then the probability that the unbiased coin was selected

Options:

- (a) $\frac{19}{21}$
- (b) $\frac{1}{3}$

(c) $\frac{1}{5}$

(d) $\frac{1}{6}$

Answer: (a)

Question: If $f(\theta) = \frac{\tan(\tan \theta) - \tan(\sin \theta)}{\tan \theta - \sin \theta}$ is continuous at $\theta = 0$, then the value of $f(\theta)$ at $\theta = 0$ is equal to

Answer: (1)

Question: Let a random variable X take values 0, 1, 2, 3 with $P(x = 0) = P(x = 1) = P(x = 2) = P(x = 3)$ and $E(X^2) = 2E(x)$. Then the value of $8p - 1$ is

Answer: (2)

Question: Let a_n be the n^{th} term of an A.P. If $S_n = a_1 + a_2 + a_3 + \dots + a_n = 700$, $a_6 = 7$ and $S_7 = 7$, then a_n is equal to

Answer: (64)

Question: Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude

such that $\frac{|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|}{|\vec{a} + \vec{b}| - |\vec{a} - \vec{b}|} = \sqrt{2} + 1$, Then $\frac{|\vec{a} + \vec{b}|^2}{|\vec{a} - \vec{b}|^2}$ is

Answer: $2(1 + \cos \theta)$

Question: Let $y = y(x)$ and $(1 + x^2) y' - 2xy = (x^4 + 2x^2 + 1) \cos x$. If

$y(0) = 1$, then $\int_{-3}^3 y(x) dx$ equals to

Options:

(a) 20

(b) 15

(c) 24

(d) 10

Answer: (c)

Question: Let the length of a latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be 10. If its eccentricity is the minimum value of the function $f(t) = t^2 + t + \frac{11}{12}$, $t \in \mathbb{R}$, then $a^2 + b^2$ is equal to

Answer: (126)

Question: If a triangle is formed using lines $y = x + 1$, $y = 4x - 8$, $y = mx + c$. If the orthocentre of this triangle is $(3, -1)$, then $m - c$ is equal to

Options:

(a) 0

(b) 1

(c) -1

(d) 2

Answer: (a)

Question: Let the foot of perpendicular from $P(5, 1, -3)$ on the line $L_1 : x - 1 = y - 2 = z$ and $L_2 : x - 2 = y = z - 1$ is Q and R , respectively. The area of triangle PQR is equal to (in square units)

Options:

- (a) $\frac{7}{2}$
- (b) $\frac{7}{\sqrt{2}}$
- (c) $\frac{7\sqrt{3}}{2}$
- (d) 7

Answer: (c)

